

Hello!
Everyone



Name: Parmar Kuldeepkumar
Pravinbhai



Enrollment no: 230610107026



Stream: Computer Engineering



College: Government Engineering
College, Palanpur



Leveraging AI and Green Skilling for Global Sustainability

A Data-Driven Approach
to Advancing UN SDG 17

Activities that I have done:

1. I calculated my ecological footprint using
“<https://www.footprintcalculator.org/home/en>”

2. I calculated carbon emission produced by
my two-wheeler vehicle and should i
replace it by E-Vehicle using
“<https://e-amrit.niti.gov.in/co2-calculator>”



Should I shift to electric vehicle?

Vehicle Segment ⓘ	2W (Two-Wheeler) ▾	Comparing Vehicle Fuel Type ⓘ	PETROL ▾
Journey Distance ⓘ	10 km	Journey Frequency ⓘ	Daily ▾
Annual Journey Distance ⓘ	3650 km	CO2 emissions factor ⓘ	39.04 g/km
Comparable EV Battery Capacity ⓘ	3 kWh	EV Range ⓘ	95 km
Battery needed for total journey ⓘ	115.26 kWh	Electricity CO2 emissions factor ⓘ	850 g/kWh
Lifecycle of the Vehicle ⓘ	15 years		

Calculate

Our Lifestyle choices, from transportation to diet, impact the climate. The CO2 emissions from our day-to-day activities have a huge impact on the climate. This is a global problem, but you can be a part of the solution. CO2 emissions Calculator helps in calculating the taillight emissions from a conventional vehicle as compared to an EV. It also calculates CO2 emissions abatement if a person switches from a conventional Vehicle to an EV.

How much will I save?

CO2 emissions reductions during the lifecycle of the vehicle 0.45 tonnes.

0.14

CO2 emissions from conventional vehicle

0.1

CO2 emissions due to electric vehicle

0.04

Annual CO2 emissions savings by opting EV

0.45

CO2 emissions reductions during the lifecycle of the vehicle

Start a new Calculation

Key Insights from Calculator Activities

- **Lessons from the Footprint Calculator:** The results showed that my lifestyle uses more resources than the Earth can naturally replace. My Earth Overshoot Day is May 14, meaning we would need 2.7 Earths if everyone lived like me. This shows that my lifestyle is not sustainable.
- **Lessons from the Carbon Emission Calculator:** By comparing different vehicles, I learned how much carbon emissions change depending on the type of transport we use. It proved that small individual switches—like choosing a 45 g/km petrol scooter over a 103 g/km car—drastically reduces daily carbon emissions.
- **Why SDG 17 (Partnerships) is My Core Focus:** The calculator activities highlight that isolated data is useless. SDG 17 is the most important goal because it connects global data knowledge, enables green technology transfers, and provides open-source digital calculators to help humanity use natural resources more responsibly.
- **The Data Center Dilemma:** While online footprint calculators are essential for global awareness, the massive cloud data centers running these complex simulations demand extreme energy and cooling water, making carbon optimization a primary tech challenge.

The En-ROADS Solution: En-ROADS is a climate simulation tool developed by MIT that helps people understand how different climate policies can affect global warming.

Sustainability: Meeting our present economic, industrial, and social needs without destroying, polluting, or compromising the core resource budget required by future generations.

Green Skilling: Training and equipping the workforce with the specific technical knowledge, capabilities, and industrial competencies needed to operate a sustainable, low-carbon economy.

SCGJ: Skill Council for Green Jobs (SCGJ) is a Government of India initiative that develops training programs and certifications to prepare people for jobs in clean energy and sustainable industries.

The SCGJ National Standard: Moving beyond simple energy generation to actively train a workforce specialized in capturing, treating, and minimizing chemical pollutants directly at the industrial source

Core Training Domains:

1. Air quality monitoring
2. Wastewater treatment
3. Safe industrial waste management
4. Renewable energy (Solar & Wind)

Pros & Cons of AI

Pros

- Helps predict solar and wind energy production.
- Improves electricity distribution and reduces energy waste.
- Monitors pollution using real-time sensor data.
- Detects pollution early so action can be taken quickly.

Cons

- Training AI models requires a lot of electricity.
- AI data centers increase energy consumption.
- AI depends on powerful computer chips
- Frequent hardware upgrades can increase electronic waste.

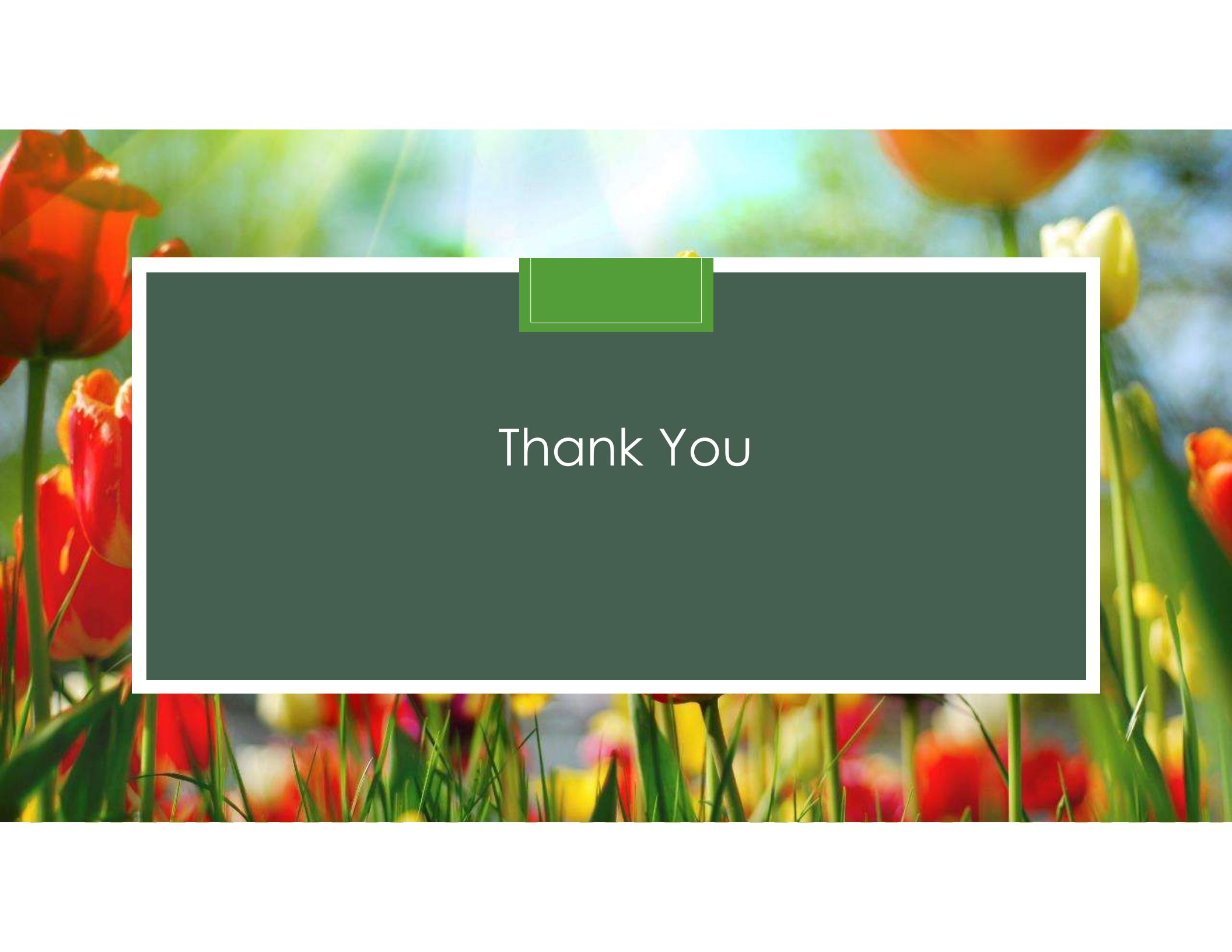
Capstone Project Ideas:

1. EcoSort – AI Computer Vision Waste Classifier

What it does: A lightweight image recognition application that uses a webcam to scan discarded trash items. The AI instantly identifies the material (e.g., plastic, paper, or electronic parts) and instructs the user on how to sort it into correct recycling or SCGJ e-waste categories.

2. Smart-Filter AI

What it does: A lightweight Python script that monitors simulated factory emission data. It automatically detects toxic chemical spikes and instantly triggers digital commands to activate secondary filtration systems before industrial pollution leaks occur.



Thank You